International Mathematics Assessments for Schools (2023-2024)

Upper Primary Division Round 2

Time: 120 minutes

Printed Name _____ *Code* _____ *Score* _____

Instructions:

- Do not open the contest booklet until you are told to do so.
- Be sure that your name and code are written on the space provided above.
- Round 2 of IMAS is composed of three parts; the total score is 100 marks.
- Questions 1 to 5 are given as a multiple-choice test. Each question has five possible options marked as A, B, C, D and E. Only one of these options is correct. After making your choice, fill in the appropriate letter in the space provided. Each correct answer is worth 4 marks. There is no penalty for an incorrect answer.
- Questions 6 to 13 are a short answer test. Only Arabic numerals are accepted; using other written text will not be honored or credited. Some questions have more than one answer, as such all answers are required to be written down in the space provided to obtain full marks. Each correct answer is worth 5 marks. There is no penalty for incorrect answers.
- Questions 14 and 15 require a detailed solution or process in which 20 marks are to be awarded to a completely written solution. Partial marks may be given to an incomplete presentation. There is no penalty for an incorrect answer.
- Use of electronic computing devices is not allowed.
- Only pencil, blue or black ball-pens may be used to write your solution or answer.
- Diagrams are not drawn to scale. They are intended as aids only.
- After the contest the invigilator will collect the contest paper.

The following area is to be filled in by the judges;
the contestants are not supposed to mark anything here.

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Total Score	Signature
Score																	
Score																	

Upper Primary Division Round 2

Questions 1 to 5, 4 marks each

1. There are x pizzas with same size in a party. If 6 children share these pizzas, then each child can eat $\frac{5}{y}$ pizzas. If 3 children share these pizzas, then each child can eat $\frac{8}{z}$ pizzas. Note that x, y and z are positive integers. How many possible values of x are there?

(A) 1 (B) 2 (C) 3 (D) 4 (E) 6

Answer: _____

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2. John takes a number and intends to multiply it with the two-digit number $\overline{ab}$, but he mistakenly multiplies it with $\overline{ba}$. If $b > a$ and the difference between the correct result and the mistaken result is 819, what is the sum of all possible values of John's number?

(A) 103 (B) 104 (C) 105 (D) 106 (E) 107

Answer: _____

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3. Two chemical substances, A and B, are composed of two basic elements, x and y. The ratios of the compositions of the two basic elements in these two substances are 5 : 3 and 1 : 2, respectively. A new substance C is made by mixing the two substances A and B in the ratio 4 : 3. In substance C, what is the ratio of the compositions of x and y?

(A) 6 : 5 (B) 5 : 6 (C) 8 : 7 (D) 7 : 8 (E) 1 : 1

Answer: _____

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4. The total number of students (girls and boys) and teachers in a school is 2024. The girls are at least as many as the teachers and the teachers are at least as many as the boys. All of them are standing in a row. It is known that there is exactly one teacher between any two consecutive girls, and there is exactly one boy between any two consecutive teachers. What is the number of teachers in the school?

(A) 675 (B) 676 (C) 677 (D) 678 (E) 679

Answer: _____

5. What are the last two digits in the decimal representation of the number $\frac{1}{5^{2024}}$?

（A）04　　（B）08　　（C）16　　（D）32　　（E）64

Answer：_____

Questions 6 to 13, 5 marks each

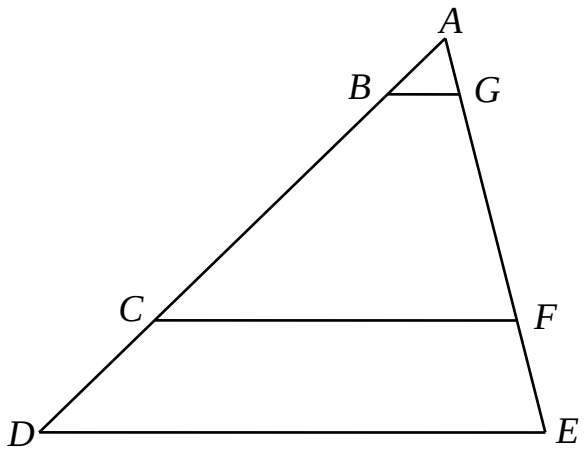
6. If 2024 is divided by a positive integer, then the quotient is the same as the remainder. What is the smallest possible divisor?

Answer：_____

7. What is the greatest positive 15-digit integer that is divisible by 88 and whose digits are 0, 1, 1, 2, 2, 2, 3, 3, 3, 3, 4, 4, 4, 4, 4?

Answer：_____

8. In triangle *ADE*, points *B*, *C* are on *AD* and *G*, *F* are on *AE* so that *BG*//*CF*//*DE* and the areas of trapezoids *BCFG* and *CDEF* are the same, as shown in the diagram below. If *BG* : *CF* = 1 : 5, what is *AC* : *BD*?



Answer：_____ : _____

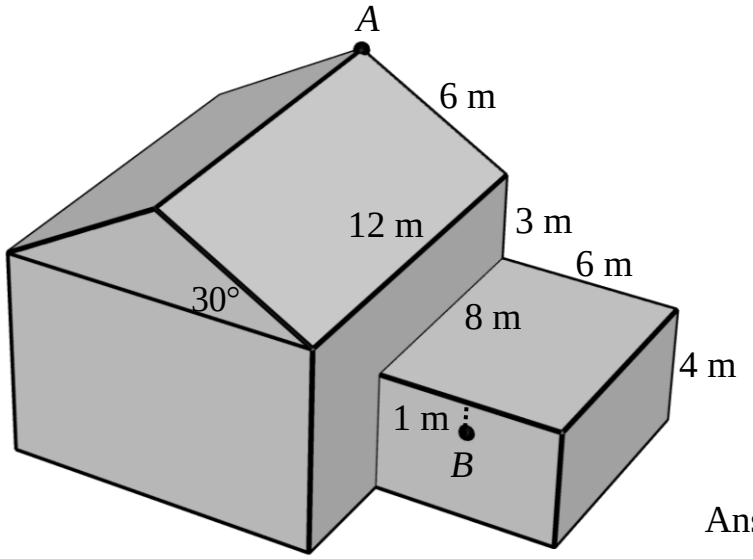
9. A positive integer is called an *IMAS* integer if it equals 24 times the sum of its digits. What is the largest possible *IMAS* integer?

Answer：_____

10. Andy, Ben and Chuck share 25 marbles. Andy wants to get at least 3 marbles and at most 10 marbles, Ben wants to get at least 2 marbles and Chuck wants to get at least 4 marbles. How many ways are there to satisfy their requirements?

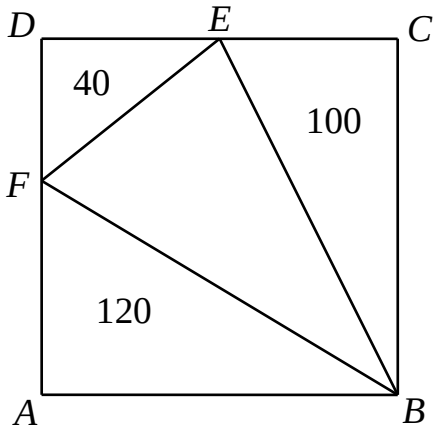
Answer：_____ ways

11. Carl has a house and he extends it for his warehouse, as shown in the diagram below. One day, he connects the point *A* with a plastic tube to point *B* which is centrally placed on the visible side of the warehouse, and is situated 1 m below the flat roof. What is the length of the shortest plastic tube, in m, required to join *A* and *B* if it is always to remain in contact with the surfaces of the building?



Answer : _____ m

12. In square *ABCD*, the areas of triangles *EDF*, *BCE* and *ABF* are 40 cm^2, 100 cm^2 and 120 cm^2, respectively, as shown in the diagram below. What is the area, in cm^2, of triangle *BEF*?



Answer : _____ cm^2

13. Two children, Ann and Bob, play a game in which an ideal coin is tossed repeatedly. The chances to obtain heads or tails on each toss of this coin are equal. Ann throws the coin 3 times, while Bob throws the coin 5 times, independently of each other. Different orders of obtaining heads and tails are regarded as different results. Thus, there are $2\times2\times2=8$ different results for Ann and $2\times2\times2\times2\times2=32$ different results for Bob, so there are totally $8\times32=256$ different toss situations that both children obtain the number of heads. How many different situations are there so that both children obtain the same number of heads?

Answer : _____ situations

Questions 14 to 15, 20 marks each
(Detailed solutions are needed for these two problems)

14. There are 10 rocks and 2 of them are radioactive meteorites. When some rocks are put into a machine, it will indicate if there is at least one radioactive meteorite among them. It will not indicate how many there are or which ones they are.
Give a way to identify the 2 radioactive meteorites by using the machine at most k times so that k is as small as possible.

Answer : $k =$ _____

15. Warren has 39 coins, 2 are gold coins and the other 37 are silver coins. Warren puts these coins in a line. Then Warren chooses the coins that satisfy one of the following conditions:
    (i) choose two silver coins so that there is exactly one gold coin between them;
    (ii) choose one gold coin and one silver coin so that the other gold coin does not lie between them;
    (iii) just choose exactly one gold coin.

    With the different positions of the two gold coins, the number of Warren's ways to choose are also different. What is the maximum number of Warren's different choosing ways?

Answer : _____ ways